# Karan Gupta

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## Experience

**Full Stack Developer & App Lead** – XCEED, NIT Jalandhar Jun 2026 – Present

*ILEED – Intelligent Learning Engagement and Entity Detection* (Node.js, Express, React, Python, MongoDB, FAISS, ONNX)

- Authored **233 PRs (219 merged)** and **740+ commits** for an AI attendance platform serving **1,000+ students** across **30+ cameras**, spanning the Node.js backend, React client, Python ML service, and mobile app.
- Integrated a FAISS vector index with the Node.js backend for live RTSP identification, and wired ONNX Runtime models into the ML service: RetinaFace detection, AdaFace IR-50 embeddings, MiniFASNetV2 anti-spoofing.
- Recovered **29 students across 9 periods** lost to DBSCAN cluster merges via per-frame voting; fixed MiniFASNet liveness preprocessing, a FAISS OpenMP deadlock, and an IVF rebuild crash past 1,000 vectors.
- Built an active-learning loop that saves only clear face crops (Laplacian sharpness, landmark yaw/pitch, detector score, runner-up margin), with automated ERP embedding generation and nightly FAISS rebuilds.
- Split the Node.js server into an IPC-routed pool of quiz and learning workers; killing one dropped **0 of 60** requests.
- Built AI Studio, generating notes and quizzes from lecture transcripts and PDF/PPTX slides (up to 600 pages) with Claude and Gemini, reserving each call's worst-case cost so credit limits can't be exceeded.
- Resolved production bottlenecks behind **11.9s** responses and an **11.3%** timeout rate, fixed an unbounded stream memory leak (heap at 91%), and cut ERP list loads from **2.0s to 915ms**.
- Shipped one-click deploys with health checks and automatic rollback, a GPU inventory dashboard with MIG/VRAM telemetry, and uptime alerting; fixed broken access control and data-leak vulnerabilities.

*XCEED Learning App – Android & iOS* (Capacitor, React, FCM, TanStack Query, GitHub Actions) | Live on Google Play

- Built the app from scratch with PIN/biometric sign-in, deep links, offline caching, and FCM push notifications.
- Designed a delta OTA update pipeline on GitHub Actions that ships only changed files (a 3-file release: **0.41 MB instead of 21.6 MB**), and automated web-to-app code sync with custom merge drivers.
- Built an SOS emergency alert that notifies every admin and shares the sender's live location every **30s**, even with the app closed, until an admin resolves it; added the iOS app (in App Store review) with Universal Links.

## Projects

**EventFlow – Distributed Workflow Orchestration Engine**
Tech: FastAPI, Python, PostgreSQL, Redis Streams, gRPC, Docker, WebSockets
GitHub: github.com/guptakaran20/EventFlow | Live: goeventflow.vercel.app

- Built a workflow engine with DAG routing, distributed Redis Streams workers, retries, and a Dead Letter Queue.
- Added fault tolerance with heartbeats, stuck-job recovery (`XPENDING/XCLAIM`), and live WebSocket updates.

**CodeArena – Real-Time Coding Battle Platform**
Tech: Next.js, Node.js, Express, MongoDB, Redis, Socket.IO, Docker, AWS EC2
GitHub: github.com/guptakaran20/CodeBattle | Live: codearenabattle.vercel.app

- Built a real-time multiplayer coding platform with battle rooms, Redis matchmaking, tournaments, leaderboards.
- Built JWT/OAuth REST APIs and Piston code execution; deployed on AWS EC2 + Nginx with Prometheus metrics.

**ImportlyAI – AI-Powered CSV Import Platform**
Tech: Next.js, TypeScript, Node.js, Express.js, Gemini API, PapaParse
GitHub: github.com/guptakaran20/ImportlyAI | Live: importlyai.vercel.app

- Engineered a batched multi-stage Gemini pipeline mapping arbitrary CSVs to a CRM schema, with 180+ tests.

## Technical Skills

**Languages:** C++, TypeScript, JavaScript, Python, SQL
**Frontend:** React.js, Next.js, Tailwind CSS, HTML5, CSS3
**Backend:** Node.js, Express.js, FastAPI, REST APIs, gRPC, WebSockets, JWT, OAuth
**Databases:** PostgreSQL, MongoDB, MySQL, Redis
**AI & ML:** Face Recognition Models, FAISS, ONNX Runtime, Gemini API, Claude API, LLM Pipelines
**DevOps & Tools:** Docker, AWS EC2, Nginx, GitHub Actions, CI/CD, Prometheus metrics, Git, Vercel
**Mobile:** Capacitor (Android, iOS), Firebase Cloud Messaging, OTA Updates

## Open Source

- **GSSoC'26:** Merged an advanced-level PR hardening auth validation and API responses in a MERN leave system.

## Education

**Dr. B. R. Ambedkar National Institute of Technology, Jalandhar**
B.Tech – Instrumentation and Control Engineering | **CGPA:** 8.49 2025 – 2029

## Achievements

- **LeetCode Knight:** 1870 rating (**top 5.7% globally**), 639 problems solved (88+ Hard), best contest rank 2,171.
- **1,890+ GitHub contributions** in the past year with a **111-day streak**; LeetCode 200 Days Badge 2026.